Statistical models

Exercise Sheet 2

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Solve the following problems before the exercise class on May 8. Use `R` for all computations and all statistical tests.

Problem 1
The data set `PlantGrowth` in `R` compares the yields of plants (described by the weight) under three different conditions. In the following, we consider analysis of variance.

a) Check if there is a difference between the three groups. Formulate the null hypothesis and the alternative and use a test with level 0.05.

b) Decide which groups are different. Formulate the test problems and use the Tukey-Kramer test with level 0.05.

Problem 2
The data set `Salaries` from the `R` package `car` provides the salaries of professors in the US. Our goal is to study how the quantities `rank` (with levels `AsstProf`, `AssocProf` and `Prof`), `discipline` (with levels `A` and `B`), `yrs.since.phd` and `sex` (with levels `Female` and `Male`) affect the so-called nine-month salary `salary`.
Consider a linear model with the variables given above. For all statistical tests it is supposed that the assumptions of a Gaussian linear model are satisfied. Use the level 0.05 for all tests and do not correct for multiple testing.

a) Estimate the intercept and the coefficients of the corresponding linear model as well as the standard deviation σ of the errors.

b) Decide which of the independent variables are relevant for the salary. Provide the null hypotheses and the alternatives and use suitable tests.

c) Discuss if the salaries of assistant and associate professors are different. Formulate the test problem and apply an appropriate test.

Problem 3
The data set `trees` in `R` provides for Black Cherry Trees the diameter, the height and the volume. Our goal is to explain the volume by the girth and the height. We expect that

$$\texttt{Volume} = c \cdot \texttt{Girth}^{\alpha_1} \cdot \texttt{Height}^{\alpha_2}$$

with $c, \alpha_1, \alpha_2 \in (0, \infty)$. By taking logarithms we obtain the linear relation

$$\ln \texttt{Volume} = \ln c + \alpha_1 \ln \texttt{Girth} + \alpha_2 \ln \texttt{Height}.$$

a) Use the least squares estimator for a linear model to determine the parameters c, α_1 and α_2.

b) Assume a Gaussian linear model. Test

$$H_0 : \alpha_1 = 2 \quad \text{and} \quad \alpha_2 = 1 \quad \text{against} \quad H_1 : \alpha_1 \neq 2 \quad \text{or} \quad \alpha_2 \neq 1$$

at level 0.01.
Hint: Rewrite the model such that you have to test if coefficients are zero.

Problem 4
For the Black Cherry Trees from Problem 3 we assume again that

$$\texttt{Volume} = c \cdot \texttt{Girth}^{\alpha_1} \cdot \texttt{Height}^{\alpha_2}$$

with $c, \alpha_1, \alpha_2 \in (0, \infty)$. Instead of studying the linear model for the logarithms, we consider non-linear regression.

a) Provide the system of equations that the non-linear least squares estimator for (c, α_1, α_2) needs to satisfy for observations (V_i, G_i, H_i), $i \in \{1, \ldots, n\}$, of volume, girth and height.

b) Estimate c, α_1 and α_2.

c) Discuss the difference between the non-linear regression model and the linear model from Problem 3. Plot the volume against the fitted values for both models.